

claude-windows / da34752f-c664-4262-9602-f5d4cfcc6db4

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\subagents\workflows\wf_774b13f4-334\agent-af49927ad008619dd.jsonl`  
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```

Transcript

1. user (2026-06-12T14:42:20.081Z)

Repo: C:\Users\avidu\Projects\badminton-highlight-indexer (branch docs/pmf-field-signals).
Review these files (read them in full):

- docs/PMF_FIELD_SIGNALS.md (NEW - hypotheses H1-H7, GREEN/AMBER/RED scoring anchors, pre-registered decision rules PROCEED/PIVOT A-C/FALSIFIED, rollup mechanics, guardrails)
- docs/FIELD_VISIT_ANSWER_SHEET.md (NEW - the per-visit capture form a non-technical field associate fills by hand)
- docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md (EXISTING, lightly edited - the recruit hand-out with the 7-question script and ground rules)

Grounding docs (also read): docs/PMF_MARKET_RESEARCH.md (esp. SS6 risks and pricing anchors), docs/NEXT_STEPS.md item 5, docs/COMMERCIALIZATION.md C13 row.

Context: the owner ratified that PMF validation happens by physically visiting Bangalore badminton academies (10-15 visits by a paid non-technical field associate) and reading signals. The kit's job: make 15 visits produce a pre-registered verdict (proceed/pivot/falsified) rather than motivated reasoning. Product = KhelSutra (rally indexing + highlights from static-camera footage; wedge = messy multi-court BYO footage; serving COGS ~\$0.30-0.60/hr via Gemini).

A critic claims the following flaw in the field-validation kit. Adversarially verify by reading the actual files - try to REFUTE it. Default to refuted if the flaw is pedantic, out of scope for a 10-15-visit directional read, or already mitigated elsewhere in the kit.

CLAIM [low] FALSIFIED row cites 'PMF section 6 risk 1' for a premise that actually lives in section 5
PMF_FIELD_SIGNALS.md line 72 says the FALSIFIED verdict means 'the "academies already record fixed-angle" premise (PMF section 6 risk 1) is wrong'. In docs/PMF_MARKET_RESEARCH.md, section 6 risk 1 is framed around AMATEUR tripod adoption ? its failure mode is 'B2C TAM collapses to coaches/clubs', i.e. academies are the refuge, not the casualty. The 'academies already record fixed-angle' premise is asserted in section 5 beachhead #1 (and the section 1 executive summary); risk 1 only touches academies via its validate clause ('observe 3-5 academies'). The citation is adjacent rather than wrong, but the pre-registration doc should point at the exact premise it would falsify. The fallback beachheads named ('#2/#3 pickleball-US, badminton Asia/Denmark') do match PMF section 5 exactly.
Proposed fix: Re-cite as 'PMF section 5 beachhead #1 premise; validation path per section 6 risk 1' (or equivalent wording).

2. user (2026-06-12T14:42:27.646Z)

```
docs/pmf-field-signals  
  M docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md  
  M docs/NEXT_STEPS.md  
?? arial.ttf  
?? artifacts/  
?? docs/FIELD_VISIT_ANSWER_SHEET.md  
?? docs/PMF_FIELD_SIGNALS.md
```

```
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardenening_status.patch
?? scratch/PR88_FIX.md
```

3. user (2026-06-12T14:42:28.682Z)

```
1      # C13 Field Signals ? what academy visits must tell us (and when to change course)
2
3      **Status:** ACTIVE ? the decision layer for the C13 beachhead validation
4      (owner-ratified direction, 2026-06-12: *PMF validation is handled by physically
5      visiting academies and reading signals on the ground*).
6      **Companions:** [`FIELD_RESEARCH_ASSOCIATE_FLYER.md`](FIELD_RESEARCH_ASSOCIATE_FLYER.md)
7      (the recruit hand-out: script + ground rules) .
8      [`FIELD_VISIT_ANSWER_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md) (the per-visit capture
9      form) .
10     [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md) (the hypotheses under test, esp. §6).
11
12     **Why this doc exists:** the flyer collects answers; this doc fixes ? **before the
13     first visit** ? what those answers *mean* and what totals trigger which decision.
14     Pre-registering the thresholds is the honesty mechanism: it stops us from reading
15     15 ambiguous conversations as whatever we hoped to hear.
16
17     ---
18
19     ## 1. The hypotheses under test (each maps to script questions)
20
21     | ID | Hypothesis (from PMF research) | Falsified looks like | Script Q |
22     |---|---|---|---|
23     | H1 | **Recording behavior exists** ? academies already produce fixed-angle / CCTV /
24     multi-court footage without us | nobody records; "we'd have to start for you" | Q1 |
25     | H2 | **Dead-footage pain** ? recorded footage piles up unwatched (the indexing wedge)
26     | footage is small, or already gets watched | Q2 ? |
27     | H3 | **An audience exists** ? coach, player, or parent would actually consume rally
28     clips/reels | shrug; "who has time to watch clips" | Q3 |
29     | H4 | **Whitespace** ? incumbents (Machaxi, VISIST, Game Theory, Stupa, BadPro+) leave
30     a gap we fit | "we use X, it's fine, fully solved" | Q4 |
31     | H5 | **Willingness to pay per match** at a price that clears unit economics | "should
32     be free" / <?100 everywhere | Q5 ? |
33     | H6 | **Multi-court chaos is a felt problem** (our moat: indexing messy BYO multi-court
34     footage) | multi-court video already useful to them | Q6 |
35     | H7 | **Behavioral commitment** ? they'll let us record on their court (Phase-2 door) |
36     polite no across the board | Q7 |
37
38     Unit-economics floor behind H5: serving COGS on the Gemini path is ~$0.30?0.60 per
39     hour of video (? ?25?50/hr; 2026-06-10 quality table). A 2?3 hr session costs
40     roughly ?75?150 to process, so **?300/match has real margin, ?100/match is
41     break-even-ish at retail** ? that's why the script's anchor ladder is ?100/300/500.
42
43     ## 2. Per-visit scoring ? GREEN / AMBER / RED anchors
44
45     Score each hypothesis on the answer sheet **from what they said and showed, not
46     what you inferred**. When between two grades, take the lower one.
47
48     | | GREEN (strong confirm) | AMBER (weak/mixed) | RED (disconfirm) |
49     |---|---|---|---|
50     | **H1 record** | shows or names actual setup (tripod/CCTV/phone rig) used ?monthly |
51     records occasionally (tournaments only) | doesn't record; no plans |
52     | **H2 dead footage** | points at real unwatched volume (?~5 hrs, "CCTV runs all day",
53     tournament backlog) | some footage, partially watched | no backlog exists |
54     | **H3 audience** | names WHO would watch and WHEN (e.g. "parents ask for clips", "I
55     review with students Mondays") | "could be useful, maybe" | no consumer identified |
56     | **H4 whitespace** | no tool today, or uses one and volunteers a gap we cover (multi-
```

court, BYO footage, price) | uses a competitor, mildly satisfied | competitor solves it, satisfied, sticky |

46 | **H5 WTP** | volunteers ??300/match unprompted, or proposes a budget framing of their own (per-student/per-month) | lands at ~?100 with hesitation | refuses to name a number / "free" |

47 | **H6 multi-court** | complains about it in their own words before/without prompting | agrees when asked | multi-court footage already works for them |

48 | **H7 Phase-2** | yes + gives a name/number/slot | "maybe, come back" | no |

49

50 **Weighting rule (behavior > words):** H7-GREEN and H2-GREEN are **behavioral** signals and count double in your gut-read; H3/H5 answers are **talk** until a Phase-2 session or a real payment corroborates them. A visit with H5-GREEN but H7-RED is a polite conversation, not a customer.

51

52

53

54

55 **Disconfirming-evidence quota:** every visit sheet must record at least one verbatim **negative** quote (the thing they liked least / wouldn't pay for). If a sheet has only positives, the visit gets re-done or discounted ? per the flyer's ground rule 4, a reasoned "no" is worth more than five vague yeses.

56

57

58

59

60 **## 3. Pre-registered decision rules (n = 10?15 academies)**

61

62 Counts, not percentages ? at this sample size we're reading direction, not estimating proportions. Evaluate after **every 5 completed sheets**; final call at the agreed n. "GREEN" below means GREEN on that hypothesis row.

63

64

65

66 | Verdict | Trigger (pre-registered) | What we then do |

67 |---|---|---|

68 | **PROCEED ? beachhead validated** | ?6 H2-GREEN **and** ?4 H5 at ?300+ **and** ?3 H7-GREEN | Run the camera pilot at the ?3 warmest academies; those contacts seed the UI-alpha Wave-1 invite list (UI_ALPHA_EXECUTION_PLAN Phase R); start the paid-pilot conversation with the single warmest |

69 | **PIVOT A ? "index the backlog" ? "record + index" service** | H2 mostly RED (<3 GREEN) but H3+H5 healthy on **fresh** recordings | The pain isn't the archive, it's that nothing useful gets made at all ? lead with the Phase-2 recording offer; camera pilot becomes the product wedge, not the demo |

70 | **PIVOT B ? partner/API motion** | H4 RED at ?5 academies (incumbents entrenched and liked) | Don't fight installed hardware head-on; explore the engine/API route through whoever owns the academy relationship (the PB Vision x PodPlay template from the 2026-06-10 audit) |

71 | **PIVOT C ? payer is the parent, not the academy** | H5 RED from owners but H3 repeatedly names parents/players asking for clips | Re-test WTP as B2C per-student reels (?/reel to parents) with the academy as channel, not customer |

72 | **FALSIFIED ? beachhead assumption fails** | <3 academies record at all (H1) **and** 0 H7-GREEN | The "academies already record fixed-angle" premise (PMF §6 risk 1) is wrong for this market ? revisit beachheads #2/#3 (pickleball-US, badminton Asia/Denmark) before building further |

73

74 Mixed outcomes that match no row: write up the per-hypothesis tallies and decide explicitly with the owner ? but **the tallies above get computed first**, so the discussion starts from the pre-registered read, not from anecdotes.

75

76

77

78 **Competitor-intel side-channel (always-on):** any visit where a competitor is named ? fill the answer sheet's competitor block (what it does, what it misses, what they pay). ?3 consistent reports about one product = update

79

80

81 ``PMF_MARKET_RESEARCH.md` §4` with field-verified rows (today's table is web-sourced).

82

83 **## 4. Mechanics**

84

85 - One `FIELD_VISIT_ANSWER_SHEET.md` per academy, same day as the visit, in the associate's words + verbatim quotes.

86

87 - The owner (or Claude session) transcribes each sheet into one row of the rollout table below (keep it in this doc or a sheet next to it); tallies and the decision-rule read-out update per 5 visits.

88

89

90 - Demo asset if a conversation earns it: the real compiled reel (`output/MahadevpuraSingles_highlights.mp4``) on a phone ? **show, never promise** (flyer ground rule 3 still applies; the associate doesn't carry the demo,

91

92

```

93     the owner does on follow-ups).
94
95     ### Rollup table (one row per completed visit)
96
97     | # | Academy | Area | Courts | Role spoken to | H1 | H2 | H3 | H4 | H5 (?) | H6 | H7 |
Gut | Phase-2? |
98     |---|---|---|---|---|---|---|---|---|---|---|---|
99     | 1 | | | | | | | | | | |
100
101     ### Tally ? decision read-out (recompute per 5 visits)
102
103     | Metric | Count | Threshold in play |
104     |---|---|---|
105     | H2 GREEN (dead footage) | /n | ?6 ? PROCEED leg; <3 ? PIVOT A leg |
106     | H5 at ?300+ | /n | ?4 ? PROCEED leg |
107     | H7 GREEN (Phase-2 yes + contact) | /n | ?3 ? PROCEED leg; 0 ? FALSIFIED leg |
108     | H4 RED (incumbent satisfied) | /n | ?5 ? PIVOT B |
109     | H1 any recording | /n | <3 ? FALSIFIED leg |
110     | H5 RED + H3 names parents | /n | recurring ? PIVOT C |
111
112     ## 5. Guardrails that bind the field work
113
114     - **Phase 1 records nothing** ? notes only (flyer rule 7).
115     - **Phase 2 consent rules are hard:** written consent from the academy AND adult
116       players; **sessions with minors are never recorded for our library** (training-pool
117       guardrail, CLAUDE.md); the academy is told plainly what the footage is for.
118     - Answer sheets contain personal contact details ? they stay between the associate
119       and the owner; transcribe to the rollup without phone numbers.
120

```

4. user (2026-06-12T14:42:36.489Z)

```

1     # Field Visit Answer Sheet ? one per academy (fill the same day)
2
3     > Print one per planned visit. Write THEIR words, not yours. A clear "no, because?"
4     > is a *good* sheet. Companion script: `FIELD_RESEARCH_ASSOCIATE_FLYER.md`.
5     > Scoring anchors (G/A/R): `PMF_FIELD_SIGNALS.md` §2.
6
7     ## Visit basics
8
9     | | |
10    |---|---|
11    | Date / time | |
12    | Academy name | |
13    | Area (neighbourhood) | |
14    | # courts in the hall | |
15    | Spoke to (name + role) | |
16    | Contact (phone/WhatsApp) ? *only if offered* | |
17    | How long the conversation ran | |
18
19    ## The seven questions ? their answers (verbatim where possible)
20
21    **Q1 ? Do they record sessions/matches today? How (phone / tripod / CCTV / hired)?**
22
23    >
24
25    **Q2 ? ? Footage sitting around that nobody watches? Roughly how much, and from what?**
26
27    >
28
29    **Q3 ? Who would watch rally clips/highlights (coach / players / parents)? When?**
30
31    >
32
33    **Q4 ? What do they use today (Machaxi / VISIST / Game Theory / BadPro+ / none)?

```

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34     What's good, what's missing, what does it cost them?**
35
36     >
37
38     **Q5 ? ? Would they pay per match for rally cuts + reel + report? What number did
39     THEY say first (?)? What was their face doing?**
40
41     >
42
43     **Q6 ? Multiple courts in frame ? does that make video useless today? Their words:**
44
45     >
46
47     **Q7 ? Open to a paid recording session on their court? (yes / maybe / no ? and who
48     to call)**
49
50     >
51
52     ## Required: the most negative thing they said (verbatim)
53
54     >
55
56     ## Best quotes (anything vivid, in their words)
57
58     >
59
60     ## Scores (circle one per row ? anchors in PMF_FIELD_SIGNALS.md $2; when torn, score
61     lower)
62     | Hypothesis | Score |
63     |---|---|
64     | H1 recording behavior | GREEN / AMBER / RED |
65     | H2 dead footage | GREEN / AMBER / RED |
66     | H3 audience | GREEN / AMBER / RED |
67     | H4 whitespace | GREEN / AMBER / RED |
68     | H5 willingness to pay ? their ? number: _____ | GREEN / AMBER / RED |
69     | H6 multi-court pain | GREEN / AMBER / RED |
70     | H7 Phase-2 door | GREEN / AMBER / RED |
71
72     ## Competitor block (only if they named one)
73
74     | | |
75     |---|---|
76     | Product | |
77     | What it does well (their words) | |
78     | What it misses (their words) | |
79     | What they pay | |
80
81     ## Your gut read (circle): COLD . POLITE . CURIOUS . EAGER
82
83     ## Phase-2 candidate? (circle): YES ? call: _____ . MAYBE . NO
84
85     ## Anything else worth knowing (layout, lighting, camera positions you noticed, who
86     really decides)
87
88     >

```

5. user (2026-06-12T14:42:36.937Z)

```

1     # Temporary Engagement: Field Research Associate (Sports-Tech, Bengaluru)
2
3     **Duration:** ~2?3 weeks, part-time (flexible hours, mostly afternoons/evenings when
4     academies are active)
5
6     **Location:** Bengaluru ? in-person visits to badminton academies

```

5 **Compensation:** ? ____ per completed academy visit + travel reimbursement *(discussed individually)*

6 **Reports to:** Avi (phone/WhatsApp: ____)

7

8 ---

9

10 ## The gig in one paragraph

11

12 We're building an AI tool that takes ordinary practice/match video from a badminton academy and automatically finds

13 every rally, builds highlight reels, and generates match reports. Before building further, we need honest, on-the-ground

14 answers from real academies about how they record, what they'd use, and what they'd pay.

15 **Your job: visit 10?15 good

16 academies, have a friendly 15?20 minute conversation with the owner/head coach using the script below, and write down

17 what they actually say.** No selling. No tech knowledge needed. If a conversation goes well, there's a Phase-2 follow-up

18 (coordinating a paid video-recording session) with additional compensation.

19 ---

20

21 ## Phase 1 ? The conversation (what you actually do)

22

23 **Who to talk to:** the academy owner or head coach ? the person who decides what the academy spends money on.

24 If you get a junior coach, be friendly, gather what you can, and ask when the owner is available.

25

26 **Opening (keep it this honest):**

27 > "Hi, I'm helping a Bengaluru engineer who's building an AI tool for badminton academies ? it takes normal practice

28 > video and automatically cuts it into rallies and highlights. He asked me to talk to a few serious academies to

29 > understand how you work with video today. Could I ask you a few questions? Takes 15 minutes, and we're not selling

30 > anything today."

31

32 **The questions (ask in this order; the starred ones matter most):**

33

34 1. Do you ever video-record sessions or matches here? How ? phone, tripod, CCTV, a hired person?

35 2. ? **Do you have recorded footage sitting around that nobody has time to watch?** (multi-court hall recordings,

36 tournament days, CCTV) ? roughly how much?

37 3. Who would watch rally clips or highlights if they existed ? coaches for training? players? parents?

38 4. What do you use today for any of this? (Listen for: Machaxi, VISIST, Game Theory, BadPro+, just YouTube/phone.)

39 What do you like / what's missing in it?

40 5. ? **If you could hand over a full session's video and get back every rally cut out + a highlight reel + a simple

41 report ? would you pay per match for that? What feels fair ? ?100, ?300, ?500 per match?** (Let THEM say a number

42 first. Note their face, not just the words.)

43 6. Your hall has multiple courts running at once, right? Does that make video less useful today?

44 7. Would you let us record one session here (we pay a small fee for the time) to show you what the tool produces on

45 YOUR court? *(This is the Phase-2 door ? just note yes/no/maybe and who to call.)*

46

47 **After each visit, fill ONE answer sheet ? the template is

48 FIELD_VISIT_ANSWER_SHEET.md

49 (print one per visit):** academy name, location, # courts, person spoken to + role + contact, answers 1?7 (their

49 words, not yours), the most negative thing they said (required), exact quotes, your gut
read, and whether they're a
50 Phase-2 candidate. *(How the answers get scored and what totals trigger which decision
lives in
51 [PMF_FIELD_SIGNALS.md])(PMF_FIELD_SIGNALS.md) ? that part is ours, not yours; just
capture honestly.)*
52
53 ---
54
55 ## Ground rules & gotchas (read twice ? this is the difference between useful and
useless)
56
57 1. **You are researching, not selling.** If they ask "how much will it cost", say
"that's exactly what he's trying to
58 figure out ? what would feel fair to you?" Never quote a price yourself.
59 2. **Past behavior beats opinions.** "Would you use video?" gets polite lies. "When did
you LAST record a session, and
60 what happened to that file?" gets the truth. Prefer the second kind of question.
61 3. **Don't promise anything.** No features, no dates, no free trials. The only thing you
may offer is the Phase-2 idea
62 (a paid recording session + free sample output) and only as "would you be open to
it".
63 4. **Write down disappointing answers verbatim.** A "no, we'd never pay for that" with a
reason is worth MORE to us
64 than five vague yeses. You're paid per completed honest visit, not per positive
answer.
65 5. **If they already use a competitor (Machaxi/VISIST/Game Theory etc.) ? goldmine.**
Ask what it does well, what it
66 doesn't, and what they pay. Spend extra time here.
67 6. **Multi-court is our favorite topic.** If they complain that video is useless because
"all the courts are in frame
68 and it's chaos" ? capture that complaint in their exact words. That's the problem we
solve.
69 7. **No recordings of people without clear permission.** In Phase 1 you record NOTHING ?
notes only. Don't photograph
70 the academy or players unless the owner offers.
71 8. **Don't oversit.** 20 minutes, thank them, leave a good impression ? we may come back
to record.
72 9. **Confidentiality:** don't describe the tech beyond the opening line, don't name
other academies' answers, and
73 don't share the answer sheets with anyone except Avi.
74
75 ---
76
77 ## Phase 2 (only for academies that said yes ? separate briefing & pay)
78
79 You coordinate a recording session: we pay the academy a small fee, set up a
phone/camera on a tripod (we provide a
80 one-page capture checklist: behind the court, elevated, whole court in frame, landscape,
1080p/60fps, don't move it),
81 record 122 sessions, and later return a sample highlight reel + report to the academy
for free. **Hard rules:**
82 written consent from the academy AND the adult players being recorded; **sessions with
minors are not recorded for
83 our library** ? adult sessions only; the academy gets told plainly what the footage is
for (building and demonstrating
84 the tool).
85
86 ---
87
88 ## Interested?
89
90 Message Avi at ___ with "FIELD RESEARCH" and your availability. First 2 visits are
paired (you + Avi or a phone
91 walkthrough) so you're never improvising.

6. user (2026-06-12T14:42:41.738Z)

```

1      # PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports
2
3      > **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year;
4      > validate live before relying on them for investment/pricing decisions. This is the
5      > market-research-of-record behind the PMF section of the living
6      > [COMMERCIALIZATION.md](COMMERCIALIZATION.md) tracker ($6). Product framing:
7      > a closed-source hosted API/SaaS (plus local mode) that ingests long-form,
8      > self-recorded, **static-camera, full-court amateur** racket-sports video and
9      > auto-detects rallies and compiles highlight reels. Primary sport: badminton;
10     > sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.
11
12     > **Confidence convention.** Figures from named industry reports, federations, or
13     > company pricing pages are tagged with source + year. Inferred items are tagged
14     > **(estimate)** / **(low confidence)**. Emerging-category market sizes vary widely
15     > between vendors; ranges are shown rather than false precision.
16
17     ---
18
19     ## 1. Executive summary
20
21     The product sits at the intersection of three favorable trends: (a) a fast-growing
22     **sports analytics market** (~USD 5?6B in 2025, ~18?22% CAGR to USD 24?36B by ~2034
23     depending on vendor); (b) a structural shift to **amateur self-recording** on action
24     cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest
25     segment); and (c) **strong racket-sport participation** ? 220M+ badminton players,
26     106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in
27     2024, +45.8% YoY).
28
29     The central tension is the **static-camera constraint**: the tool works best on fixed,
30     full-court amateur footage and *fails* on broadcast/panning footage. This is both a
31     moat (most competitors target team sports / broadcast) and a filter on the addressable
32     user ? the customer must already set a camera on a tripod behind the court. That
33     behavior is concentrated in **coaches/academies, clubs, and serious enthusiasts**, not
34     the median casual player.
35
36     **Recommended beachhead (strongest single):** **badminton coaching academies and
37     serious club/competitive players in India** ? a large, rapidly formalizing academy
38     ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing
39     willingness to pay for coaching, an underserved tooling gap (incumbents are
40     team-sports/broadcast), and a direct-but-immature competitor field. The product is
41     *native badminton-first* ? a genuine edge here. Secondary: **(2) US/global pickleball
42     players & clubs** (capital-rich, fastest-growing, but SwingVision already strong);
43     **(3) badminton academies/clubs in East/SE Asia + Denmark** (highest density).
44
45     **Pricing** clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo)
46     and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A
47     software-only BYO-camera model undercuts the hardware players; realistic anchors
48     **(estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr per location.**
49
50     **Top risks:** (1) amateurs won't reliably set up a tripod ? TAM collapses to
51     coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality
52     variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia,
53     China) are the most price-sensitive and hardest to bill at Western SaaS prices.
54
55     ---
56
57     ## 2. Market sizing & trends
58
59     ### 2.1 Sports analytics / sports-video market
60     - **2025 size:** ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business
61       Insights); other vendors USD 5.6?6.0B. *(2025)*

```


62 - **Growth:** CAGR 15.7%?22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI
63 ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%);
64 Future Market Insights ? ~USD 36.2B by 2035 (22.1%). *(reports 2024?2025)*
65 - **Video analytics is the leading segment** within sports analytics (Grand View /
66 Fortune BI) ? the directly relevant sub-market.
67 - **Interpretation (estimate):** the **amateur/club auto-highlight** slice is a small but
68 fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The
69 product targets the long tail the big numbers under-serve.
70
71 **2.2 Action-camera / amateur recording adoption**
72 - **2024 size:** ~USD 6.5B (GMIInsights) ? ~USD 16.9B by 2034 (~10% CAGR). *(2024?2025)*
73 - **Sports is the largest end-use segment** of action cameras (GMIInsights, 2024).
74 - **GoPro ~47% share, DJI ~20%** (electroIQ/industry, 2024); phones + cheap tripods are
75 the de-facto amateur rig.
76 - **Implication:** the capture-side behavior the product depends on is already
77 widespread and growing ? **for the segments that record** ? validating "BYO-camera,
78 software-only."
79
80 **2.3 Participation & popularity by sport and geography**
81
82 | Sport | Participation signal | Source (year) |
83 |---|---|---|
84 | **Badminton** | 220M+ play regularly; 160+ countries | BWF est. via BadmintonBites
(2023) |
85 | Badminton ? China | ~80M registered, up to ~100M (largest base) | industry
compilations (2023) |
86 | Badminton ? India | 20M+ recreational by 2023; academy boom + Khelo India | industry
compilations (2023) |
87 | Badminton ? Indonesia | National sport; ~1M club players | industry (est.) |
88 | Badminton ? Denmark | 100,000+ registered club members (Europe #1) | industry (2023) |
89 | Badminton ? Japan/Korea | Major school-club + workplace sport | industry (2023) |
90 | **Tennis** | **106M players (2024)**, +25.6% over 5 yrs (87M in 2021) | ITF Global
Tennis Report (2021, 2024) |
91 | Tennis ? Asia | ~1/3 of players (33M of 87M) | ITF (2021) |
92 | **Pickleball ? US** | **19.8M players (2024)**, +45.8% YoY, +311% over 3 yrs; ~6.2M
core | SFIA / Pickleheads / The Dink (2024) |
93 | Pickleball ? US courts | ~70,000 courts, +55% places to play YoY | SFIA / Pickleheads
(2024) |
94 | Pickleball ? global | IFP 78 member countries; UK +73% in 2024; ~282M played monthly
per PPA claim **(low confidence)** | various (2024?2025) |
95 | **Table tennis** | 30M+ players; 227 member associations | ITTF (current) |
96 | **Cricket** | 2.5B+ fans; 500M+ in India (mostly **fans**, not self-recorders) | Cricket
Rise / Nielsen (2024?2025) |
97
98 - **Pickleball market value:** ~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR
99 (Market.us/Custom Market Insights, 2024?2025).
100 - **Badminton equipment market:** ~USD 7.23B global (2023), Asia-Pacific dominant.
101
102 **Geographic read:** badminton density is overwhelmingly Asian + Denmark; tennis is
103 broad/high-income; pickleball is US-dominant, capital-rich, globalizing fast; cricket
104 is huge as **fandom** but its amateur self-recording + static-full-court fit is weak
105 (large field, not a fixed-frame court). For **this product's static-camera fit**,
106 badminton/pickleball/tennis/table-tennis are the core; cricket is a long-tail roadmap
107 item, not a beachhead.
108
109 ---
110
111 **3. Customer segment evaluation**
112
113 | Segment | Market-size signal | Willingness to pay | Accessibility | Static-camera fit
| B2C/B2B | Verdict |
114 |---|---|---|---|---|---|
115 | **Individual amateur/enthusiast** | Largest raw TAM (100s of millions) | **Low** ?
competes with free; ~USD 5?15/mo ceiling | High via app stores; high CAC/churn | **Mixed** ?
most won't set a tripod | B2C | Volume play, weak economics; use as funnel |

116 | ****Coaches & coaching academies**** | India academy boom; thousands across Asia |
****High**** ? already sell coaching; USD 300?1,500/yr plausible | Reachable via
federations/distributors/social | ****Strong**** ? already film fixed-angle | B2B/B2B2C | ****Best
fit**** ? buyer, recorder, payer aligned |

117 | ****Clubs & leagues**** | Denmark 100k+; thousands of clubs Asia/EU | Medium-high; USD
500?3,000/yr | Medium ? committee sales cycle | ****Strong**** | B2B/B2B2C | Strong secondary;
longer cycle |

118 | ****Tournament/event organizers**** | Many small/regional events | Medium ? per-
event/seasonal | Medium ? episodic | ****Strong**** | B2B | Good per-event upsell; lumpy |

119 | ****Schools & universities**** | Large (esp. Asia/Japan clubs) | Low-medium ? budget-
constrained | Low ? slow procurement | Strong | B2B | Slow; grant/Khelo-India dependent |

120 | ****National/regional federations**** | Few, high-value | Medium-high; procurement-
heavy/political | Low ? long cycles | Strong | B2B | Lighthouse/credibility, not beachhead |

121 | ****Broadcasters / streamers**** | High value per deal | High | Low ? incumbents
entrenched | ****Poor** ? panning = failure mode** | B2B | ****Avoid**** ? violates core constraint |

122

123 ****Key dynamics:**** the static-camera constraint is a ****B2B filter**** ? coaches/academies/
124 clubs already record from a fixed angle, so the product fits existing behavior (cleanest
125 PMF). B2C is a large low-WTP, high-CAC funnel best used to feed B2B. Broadcasters
126 explicitly violate the constraint ? exclude.

127

128 ---

129

130 **## 4. Competitive & pricing benchmarks**

131

132	Product	Category	Model	Price	Notes (year)
133	---	---	---	---	---
134	**Veo (Cam 3)**	Team-sports auto-filming	Hardware + sub	Camera ~USD 1,533 (5G ~1,886) + ~£395?895/yr	15,000+ clubs/80 countries; \$115M raised (2025)
135	**Trace**	Soccer auto-filming	**Free camera + sub**	USD 180/yr (Basic) ? 300 (Pro) Camera leased free with sub	(2025)
136	**Pixellot (Air)**	Auto camera/streaming	Hardware + sub	Camera ~USD 949 + USD 69?167/mo	Institutional/league; 10,000+ clubs (2025)
137	**Spiideo**	Auto-filming/analytics	Sub incl. camera	~USD 3,800/yr (1 team) ? 6,100/yr (club)	Institutional budgets (2025)
138	**Hudl**	Team video analysis	SaaS tiers	From ~USD 400/yr	Broad team-sports incumbent (2026)
139	**SwingVision**	**Tennis/pickleball AI, single iPhone**	Software freemium	**USD 179.99/yr** ; free 2 hrs/mo	Closest analog; 200k+ monthly users; \$10M+ raised (2025)
140	**BadPro+**	**Badminton AI on phone**	**Pay-per-use, no sub**	Per-analysis (? "< one coaching session")	Direct badminton competitor; shot detection, heatmaps, auto-highlights (2025)
141	Niche pickleball/badminton apps Various Mixed Varies Fragmented; few true auto-highlight tools (2025)				
142					
143	**What buyers actually pay (synthesis):** consumer software-only anchors at ~USD				
144	100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club,				
145	USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses *pay-per-use* ?				
146	signaling a real badminton AI market but **unproven subscription WTP** (opportunity +				
147	warning). **Positioning:** software-only + BYO-camera undercuts hardware players and				
148	sells worldwide via SaaS. Realistic anchors **(estimate): B2C USD 60?180/yr; B2B				
149	academy/club USD 300?1,500/yr** per location. The differentiators (cheap offline				
150	self-improving detection, per-rally stats/tagging, clean licensing) support a **B2B**				
151	value story more than a B2C one.				
152					
153	---				
154					
155	## 5. Recommended beachhead				
156					
157	Ranked by **WTP × participation density × existing recording behavior × static-camera 158 fit × competitive whitespace.**				
159					
160	### #1 (strongest) ? Badminton coaching academies & serious players, **India**				
161	20M+ recreational players (2023); rapidly **formalizing academy ecosystem** + Khelo				
162	India tailwind; academies already record fixed-angle; coaches are buyers who monetize				

163 instruction (high relative WTP); incumbents ignore badminton; the only direct badminton
164 AI (BadPro+) is early with unproven pay-per-use. Product is *native badminton-first*.
165 **Watch-outs:** price sensitivity (need INR-tuned tiers); sell to the **academy**, not
166 the individual.
167

168 ### #2 ? Serious pickleball players & clubs, **US (then UK/Australia/Asia)**
169 Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts;
170 strong recording/sharing culture; pickleball is on the roadmap. **Watch-out:**
171 **SwingVision already owns single-camera tennis+pickleball AI** at USD 179.99/yr (200k+
172 monthly users) ? head-to-head differentiation, not greenfield.
173

174 ### #3 ? Badminton academies/clubs, **East/SE Asia (China, Indonesia, Malaysia, Japan,
Korea) + Denmark**
175 Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea
176 school-club & workplace systems), mature club infra, fixed-court recording.
177 **Watch-outs:** China needs local hosting/partners + is regulatory-complex; SE Asia
178 price-sensitive; Denmark/EU small but high-WTP and easy to pilot.
179

180 **Bottom line:** lead with **India badminton academies** (whitespace + behavior + buyer
181 alignment), use **enthusiast B2C as the funnel**, validate a **club/academy B2B2C
182 motion**, then expand to **pickleball-US** (out-build SwingVision) and
183 **badminton-Asia/Denmark** (density).
184

185 ---
186

187 ## 6. Risks & open questions (what would invalidate the thesis)
188

- 189 1. **Tripod/static-camera adoption among amateurs.** If serious amateurs won't set up a
190 fixed camera, B2C TAM collapses to coaches/clubs. *Validate:* observe 3?5 academies +
191 50?100 enthusiasts on whether they already record fixed-angle.
- 192 2. **Free / near-free alternatives** (YouTube, free app tiers, BadPro+ pay-per-use) cap
193 B2C WTP. *Validate:* fake-door pricing at USD 60?180/yr.
- 194 3. **Capture-quality variance = the failure mode** (gym lighting, off-center angle,
195 occlusion, mixed resolutions). *Validate:* benchmark rally-detection P/R on 100+ real
196 amateur clips; define a "good-capture" spec + auto-reject (ties to
197 `VIDEO\_RECORDING\_GUIDELINES.md` + the observability reports).
- 198 4. **Monetization mismatch in best-participation markets** (India/Indonesia/China are
199 price-sensitive, hard to bill). *Validate:* local pricing/payments pilot; per-academy
200 INR licensing.
- 201 5. **Competitive response** (SwingVision or a team-sports incumbent adds racket sports).
202 *Validate:* ship a badminton-quality advantage incumbents can't trivially match
203 (offline self-improving detection, per-rally tagging).
- 204 6. **B2B sales-cycle drag** for clubs/federations/schools. *Validate:* single-coach/
205 academy self-serve before institutional deals.
- 206 7. **Clean-licensing positioning** matters mainly to B2B buyers ? confirm it's a real
207 purchasing criterion via 5?10 buyer interviews.
208

209 **Cheapest next steps:** (a) 10?15 discovery interviews with Indian badminton academy
210 owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door
211 pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).
212

213 ---
214

215 ## 7. Sources
216

- 217 - Sports analytics market: Precedence Research; Fortune Business Insights; Grand View
218 Research; MarketsandMarkets; Future Market Insights. *(2024?2025)*
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- 223 - <https://www.futuremarketinsights.com/reports/sports-analytics-market>
- 224 - Action camera market: [https://www.gminsights.com/industry-analysis/action-camera-](https://www.gminsights.com/industry-analysis/action-camera-market)
225 market (2024);
<https://www.precedenceresearch.com/action-camera-market> (2024);

226 <https://electroi.com/stats/action-camera-statistics/> (2024)
 227 - Badminton participation/market: <https://badmintonbites.com/how-many-badminton-players-are-there-in-the-world/> ;
 228 <https://stringsandpaddles.com/where-is-badminton-played-10-countries-where-its-popular/> ;
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 231 <https://www.thesportsschool.com/top-10-badminton-academies-in-india/>
 232 - Tennis: <https://www.itftennis.com/en/news-and-media/articles/itf-global-tennis-report-participation-hits-106-million-in-five-years/> ;
 233 <https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-gains/>
 234 - Pickleball: <https://sfia.org/resources/as-pickleball-continues-unprecedented-growth-in-every-age-group-and-region-for-third-straight-year-significant-investments-still-needed-for-court-and-facility-demand/> ;
 235 <https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-nearly-20-million-in-2024/> ;
 236 <https://www.pickleheads.com/guides/pickleball-statistics> ;
 237 <https://www.custommarketinsights.com/report/pickleball-market/> ;
<https://market.us/report/pickleball-market/>
 238 - Table tennis: https://en.wikipedia.org/wiki/International_Table_Tennis_Federation
 239 - Cricket: <https://cricketrise.com/cricket-fans-in-india/> ;
 240 <https://www.nielsen.com/insights/2025/global-cricket-connecting-fans-driving-roi/>
 241 - Competitors & pricing: <https://www.véo.com/en-us/pricing> ; <https://traceup.com/pricing>
 ;
 242 <https://www.pixellot.tv/> ; <https://www.hudl.com/pricing> ;
<https://swing.vision/subscribe> ;
 243 <https://www.badproplusapp.com/>
 244

7. user (2026-06-12T14:42:47.652Z)

48- (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with
 49- paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
 50-3. **M0.4 next increment (\$0\local):** swap the Gen-0 LogReg for **ActionFormer/ASFormer** (MIT, minutes on the
 51- 2070S ? still lightweight, NOT the heavyweight Gen-2 stack) inside the existing harness;
 per-video threshold
 52- calibration (the LOVO-vs-oracle gap is visible per fold in the harness output).
 53-4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`);
 merged-prototype LOVO across
 54- badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated
 55- by C3 (provenance multiplies per sport).
 56:5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions:
 57- "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies
 58- (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:
 59- `output/MahadevpuraSingles\_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner ratified the
 60- physical-visit approach):** `docs/FIELD\_RESEARCH\_ASSOCIATE\_FLYER.md` (recruit hand-out: script, ground rules,
 61- Phase-2 consent; owner fills ?/contact placeholders) + `docs/FIELD\_VISIT\_ANSWER\_SHEET.md` (per-visit capture
 62- form) + `docs/PMF\_FIELD\_SIGNALS.md` (G/A/R scoring anchors + **pre-registered decision rules** ? PROCEED /
 63- PIVOT A-C / FALSIFIED thresholds fixed before the first visit).
 64-6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified working; confirm old-key

8. user (2026-06-12T14:42:48.495Z)

34-| | **? PENDING (needs action / not started)** | | | | |

35-| C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 + AAC** (`libx264` in `backend\pipeline\stitcher\compiler.py`) ? encode-side AVC patent exposure remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4\L40)** for HW encode. See §4. Decision: provision Ada-class GPU? |

36-| C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic `setup.py` shadows the build system. | Design in `DEFERRED\_HARDENING\_PLAN.md` (#7) ? **awaiting your approval** before code. |

37-| C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process` blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in `PLATFORM\_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice = `DEFERRED\_HARDENING\_PLAN.md` (#16). |

38-| | **? IN-FLIGHT (in progress / watch)** | | | | |

39-| C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **(C) Gemini 2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make "provider registry + paid fallback" a core principle; build the owned model **after** the scaffolding/flywheel are solid. See `PLATFORM\_ARCHITECTURE.md` §5A. |

40-| C13 | **Beachhead validation** | **P0** | ? | Market research (`PMF\_MARKET\_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead. Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R benchmark on 100+ real amateur clips; fake-door pricing (B2C ~\$99/yr, B2B academy ~\$600/yr). See §6. |

41-| C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)** selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). | Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See `GOLDEN\_SET\_VENDORING.md`. |

42-[Omitted long context line]

43-[Omitted long context line]

44-| C6 | **Inbound HEVC decode** | **P2** | ? ?? | Decoding users' GoPro H.264/**HEVC** uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual; confirm decode-side exposure with counsel before launch. |

45-| C7 | **Hosting unit-economics** | **P2** | ? | Per-video cost is *cheap* for Gemini (~\$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured. | **Benchmark \$/video & latency on the target VM** for A/B/C before pricing. See §5. |

46-| C10 | **Value layer (product surface)** | **P2** | ? | Minimal exports shipped (EDL\JSON\CSV + `basic\_stats`, PR #51). | Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6. |

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198-

199----

200-

201-## Changelog

202-

203-- **2026-06-07** ? Added the market-research-of-record (`PMF\_MARKET\_RESEARCH.md`):

204: ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13

205- "beachhead validation" workstream. Wired the summary into §6.

206-- **2026-06-07** ? Created this living tracker. Merged the 2026-05-30 monetization

207- audit + the commercial-readiness review. **Refreshed since the audit:** C1 AGPL

208- removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,

209- the audit's AV1 recommendation is not yet implemented); C9 packaging tied to

210- `DEFERRED\_HARDENING\_PLAN.md`; reflected PR #50/#51 (input-quality reports, value-layer

9. user (2026-06-12T14:44:29.951Z)

Structured output provided successfully

10. assistant (2026-06-12T14:44:35.851Z)

Verification complete. The claim is refuted as pedantic: all its factual observations are accurate, but the citation is a defensible risk-register pointer (§6 risk 1's validate clause is exactly the observation the FALSIFIED trigger operationalizes), the falsified premise is quoted verbatim in the row itself, and neither the pre-registered trigger nor the consequent action is affected.